

① Initial Speed $u = 20 \text{ m/s}$

$$\theta = 37^\circ$$

$$g = 10 \text{ m/s}^2$$

$$\sin 37^\circ = \frac{3}{5}, \quad \cos 37^\circ = \frac{4}{5}$$

$$u_x = u \cos \theta = 20 \left(\frac{4}{5} \right) = 16 \text{ m/s}$$

$$u_y = u \sin \theta = 20 \left(\frac{3}{5} \right) = 12 \text{ m/s}$$

a) Time of flight

$$T = \frac{2u_y}{g} = \frac{2(12)}{10} = \underline{\underline{2.4 \text{ s}}}$$

b) Horizontal range

$$R = v_x T = 16(2.4) = \underline{\underline{38.4 \text{ m}}}$$

c) Maximum height

$$H_{\text{max}} = \frac{u_y^2}{2g} = \frac{12^2}{20} = \underline{\underline{7.2 \text{ m}}}$$